Literature Review

Sleep Apnea Diagnose Support in Children

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Abstract

Sleep plays a decisive role in our physical and mental health. Any sleep disorder at an age when cognitive and emotional development is in higher demand can have significant impacts [6]. However, the accuracy and correct timing of diagnosing such disorders can be challenging, especially in pediatric populations, often due to resource-intensive and time-consuming methods like polysomnography and the high demand for specialists.

Machine learning has shown new routes for extreme improvements in performance and accessibility. By synthesising recent studies, we will highlight a deeper and more comprehensive understanding of how ML can support experts in obtaining more substantial confidence levels in diagnosing sleep disorders with a special focus on obstructive sleep apnea OSA.

Keywords: Sleep, apnea, disorders, polysomnography, children, diagnose, diagnosis, machine learning, ML.

1. Introduction

Sleep is fundamental to healthy development in children and adolescents. However, sleep disorders such as sleep apnea are prevalent in this population segment, affecting approximately 30% of children worldwide [13, 21].

Early and accurate diagnosis is crucial to mitigate the risk of development complications like cognitive deficits and behavioural or cardiovascular issues. On-time and accurate diagnosis is essential to mitigate these risks, yet gold-standard methods, such as polysomnography, still show significant challenges due to limited resources, high costs, confort levels, availability, specialised infrastructure and report analysis.

Digital technology and artificial intelligence advancements, especially in machine learning (ML) that continous to show potential to decrease existing constraints and support health professionals when diagnosing pediatric sleep disorders. Applications range from automating data analysis to wearable devices for home usage and real-time monitoring.

The following review aims to provide an in-depth examination of what the literature can underline, from the traditional methods to the role of machine learning in supporting the diagnosis of sleep disorders among children aged from 0 to 18. Will approach the current landscape of sleep disorder diagnosis, the different obstacles or constraints of the traditional methods and where ML can help bridge those gaps.

By synthesising the findings in recent literature, we aim to bring the challenges and opportunities in this field into the spotlight, ultimately guiding future research toward clinically valid solutions.

2. Background

2.1. Sleep Apnea

Sleep apnea is one of children's most consistent sleep disorders, defined by repeated breathing interruptions during sleep. [8] It disrupts the patient's sleep cycle, preventing a healthy restoration of both body and mind. The cumulative effects of sleep deprivation can manifest in various problems, from poor academic performance and irritability to cardiovascular strain.

We can see a high contrast in symptoms between adults and children. Adults usually report high fatigue levels during the day, while children may exhibit hyperactivity, behavioural problems, or even developmental delays—challenging differences when diagnosing such conditions as symptoms are frequently mistaken with other conditions, such as attention deficit or hyperactivity (ADHD). [23]

It affects 1-5% of children [12, 21] and is strongly linked to obesity, enlarged tonsils, and craniofacial abnormalities. [5] The above reinforces the need to combine objective data with a specialised clinical assessment. Current diagnostic procedures primarily rely on polysomnography exams, the gold standard for detecting obstructive sleep apnea (OSA), relying on breathing patterns, oxygen levels and brain activity monitoring and records. Despite its accuracy, reliance on this method limits accessibility and delays diagnosis.

2.2. Impacts on Cognitive Development

Sleep apnea can significantly impair cognitive development in children. Different studies show that sleep is vital for memory consolidation [15,16], emotional regulation and overall brain development [3,17]. Chronic disturbances of these processes translate into far-reaching implications. Untreated children with this disorder often struggle with attention span, emotional stability and academic performance due to disrupted rapid eye movement (REM), memory impairments can constrain learning new information and retaining previously learned materials.[22,7]

2.3. Diagnosing & Machine Learning Support

A pediatric sleep disorder diagnosis procedure traditionally involves a comprehensive patient history, physical examinations, and specialised questionnaires.

Studies refer to this method as Sleep Clinical Record (SCR) [1]. This clinician observation and analysis is key to considering aspects that may introduce or camouflage the disorder, for example, anatomical abnormalities such as enlarged adenoids or tonsils can obstruct the airway, directly contributing to the development of obstructive sleep apnea (OSA) [20].

Additionally, craniofacial structure variations, including smaller jaw sizes or high-arched palates, can exacerbate airway blockages. Conversely, the body's natural compensatory mechanisms, such as increased respiratory effort, can temporarily mitigate symptoms, making the condition more challenging to detect during initial assessments [5].

While informative, these methods can become subjective and vary significantly among sleep health experts. The first challenge starts with a child's limited ability to express symptoms and reliance on parental observations. The process can take several iterations and different approaches until the clinician defines the treatment method that best fits a patient. However, studies have shown that clinical assessments alone may not reliably separate primary snoring from obstructive sleep apnea in children [2].

Professionals rely on polysomnography, the gold standard for diagnosing OSA, as it provides detailed information on sleep architecture and respiratory events [4]. Bringing this in-depth understanding of the patient's sleep and biometric levels reinforces the clinician's confidence, as research shows that a traditional SCR variation rate may not replace a PSG entirely besides being an expensive and complex procedure available for the minority. [7]

PSG can be complex due to its level of electronic dependency. The procedure relies on a minimum of twelve channel records, requiring at least twenty-two wire attachments to the patient. These channels may vary from lab to lab and might be adapted to clinicians' needs.

This sleep study is mainly conducted at the hospital or health facilities, where the child needs to stay overnight, out of his comfort zone, to be continuously monitored by a credentialed technologist. It records the physiological changes that occur during the night sleep. The captured metrics are basic brain activity through Electroencephalography (EEG), eye movement from Electrooculography (EOG), muscle or skeletal muscle activation with Electromyography (EMG), and heart rhythm with an Electrocardiography (ECG). [19]

Figure 1 -Polysomnography, Child [14]

The exam level of complexity adds a clear challenge when monitoring children at younger ages, who can quickly become impatient and struggle to remain still for more extended periods of time and keep the electrodes attached for a proper reading.

Clinicians need to combine their own observation process with PSG results for better accuracy before a final diagnosis, as basing it only on one of those methods can easily lead to the wrong interpretation of the symptoms and wrong treatment, which can introduce an array of new problems to the patient.

Even the most experienced professionals sometimes need more than one intervention until a well-adjusted treatment after a diagnosis. The assessment based on the patient's symptoms, history and anatomy crossed with PSG results are mostly based on his experience. This is where technology can assist even more of these clinicians. The application of machine learning algorithms on top of this captured data can help these professionals gain more conviction when diagnosing and defining specific treatments.

3. Methodology

This article uses a mixed systematic literature review methodology supported by Design Science Research Methodology (DSRM) and PRISMA (Preferred Reporting Items for

Systematic Reviews and Meta-Analyses). By following a PRISMA approach, the research can remain grounded in existing knowledge and highlight its novelty. Meanwhile, DSRM will provide the needed systematic iteration between developing, testing, and refining until an innovative solution to the identified problem is found.

To complement the literature review, interviews were conducted with key clinical experts to provide contextual and practical insights. Joana Marques, a PhD nurse specialising in pediatric sleep disorders and author of the podcast "Sono Trocado por Miúdos", facilitated introductions to two physicians from different hospitals specialising in sleep medicine, Dr. Rosário Trindade Ferreira from Santa Maria University Hospital and Dr. Helena Loureiro from "Hospital da Luz". These experts provided vital guidance on diagnostic workflows, challenges in current methodologies like how polysomnography is adjusted to the country's reality and the practical implications of integrating machine learning into clinical practice. A document search was conducted using SCOPUS to reinforce the provided guidelines and resources.

3.1 Questions

Q1. What are the traditional diagnosis methods and processes for sleep apnea in children?
Q2. What elements strongly depend on clinicians' analysis of those diagnosis methods?
Q3. Can a sleep apnea diagnosis rely only on clinical records, or is the combination of polysomnography necessary for better accuracy?
Q4. Can ML be applied to support health experts confidence levels when diagnosing a OSA to a child?

3.2 PRISMA for Systematic Review

The PRISMA framework was utilized to identify, screen, and synthesize literature relevant to pediatric sleep apnea diagnosis and machine learning applications.

3.2.1 Search Strategy

Searches were conducted using Scopus and Web Of Science using the target keywords.

3.2.2 Inclusion and Exclusion Criteria

Table 1 - Search inclusion and exclusions criterias

Inclusion Criteria	Observations
Age interval from 0 to 18	The study is focused on the pediatric population.
Sleep Apnea	There are other sleep disorders, like
Diagnosis Methods	Focus on support diagnosis; this highlights the need to know the different diagnosis methods.
Combination of SCR and PSG.	The end goal of this research is to analyse how the application of ML algorithms can support diagnosis when considering patient clinical history and PSG reports.
References to the application of machine learning algorithms.	Its important to have a baseline for conducted studies of the application of machine learning to support OSA diagnosis.
Studies made in the last 10 years.	Even though the diagnosis methods approached have been the gold standard for almost twenty years, reducing this

	time interval when interacting with ML applications while still having some buffer is important.
Studies written in English or Portuguese.	My research will be based on Portuguese data, which may limit this review. Also, studies for OSA diagnosis are usually based on universal data and constraints, where the application of the different methods may vary.

Exclusion Criteria	Observations
Ages > 18	The study focuses on ages with more significant cognitive development.
Other sleep disorders that not OSA.	To keep this research focused and the timeframe realistic, all other disorders, such as insomnia, Restless legs syndrome, night terrors, and many more, are not considered. However, this does not mean that the final results cannot be adapted for any of those disorders.
Before 2014	The considered interval is 2014 > 2025.
There is no reference to SCR or PSG methods	To improve our diagnosis performance, we need to consider both clinical analyses and PSG reports.
Specific conditions analysis.	Several conducted studies cover OSA and PSG as diagnosis methods for target patients suffering from specific conditions, such as Down syndrome or birth malformations. This research focuses on prognosis based on the captured and used data that experts can adjust based on those special conditions.

3.2.3 Queries

Scopus

Query 1

TITLE-ABS-KEY (sleep AND apnea AND children AND polysomnography AND diagnose OR diagnosis) AND PUBYEAR > 2013 AND PUBYEAR < 2026 AND (LIMIT-TO (DOCTYPE , "re") OR LIMIT-TO (DOCTYPE , "ar") OR LIMIT-TO (DOCTYPE , "cp")) AND (LIMIT-TO (LANGUAGE , "English") OR LIMIT-TO (LANGUAGE , "Portuguese"))

Query 2

TITLE-ABS-KEY (sleep AND apnea AND children AND polysomnography AND diagnose OR diagnosis AND ml OR machine AND learning) AND PUBYEAR > 2013 AND PUBYEAR < 2026 AND (LIMIT-TO (DOCTYPE , "re") OR LIMIT-TO (DOCTYPE , "ar") OR LIMIT-TO (DOCTYPE , "cp")) AND (LIMIT-TO (LANGUAGE , "English") OR LIMIT-TO (LANGUAGE , "Portuguese"))

Web Of Science

TS=(sleep AND apnea AND children AND polysomnography AND machine AND learning AND diagnose OR diagnosis)

3.2.4 Results & Extration

Scopus

This search filtered 723 results based on articles, review articles and papers in English or Portuguese at the time interval between 2014 and 2025.

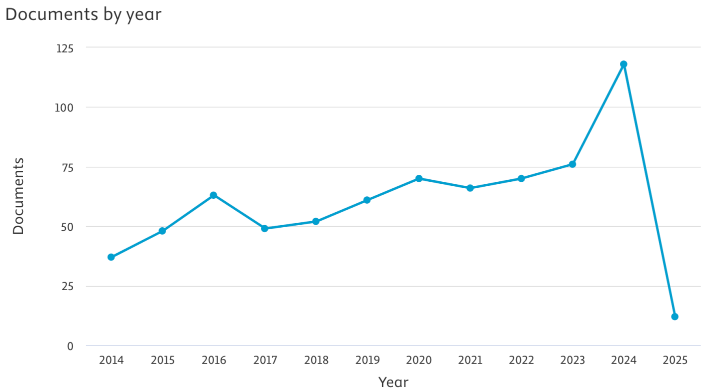


Figure 2 - Scopus documents year overview

We can see an exponential increase in published documents in 2024, with 118 documents found, which might reflect a recent focus on the subject.

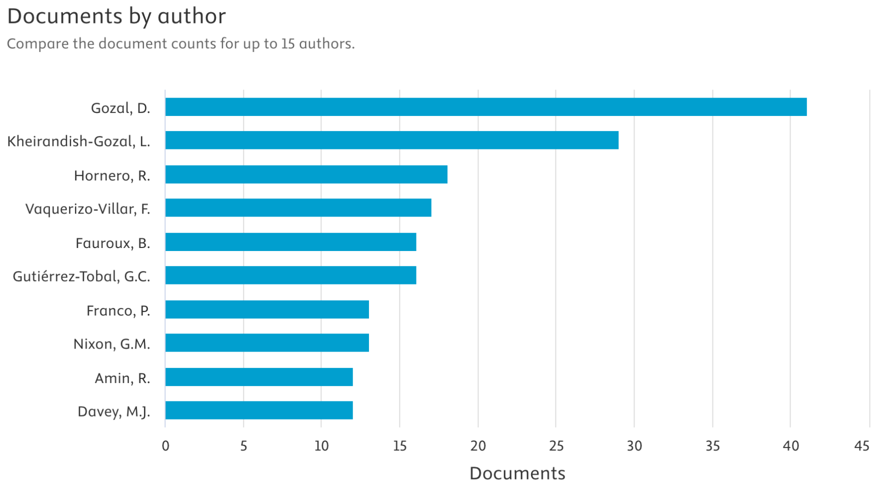


Figure 3 - Scopus documents authors overview

Professor David Gozal, MD, MBA, PhD with more than 40 documents on this search was a clear influence. [18]

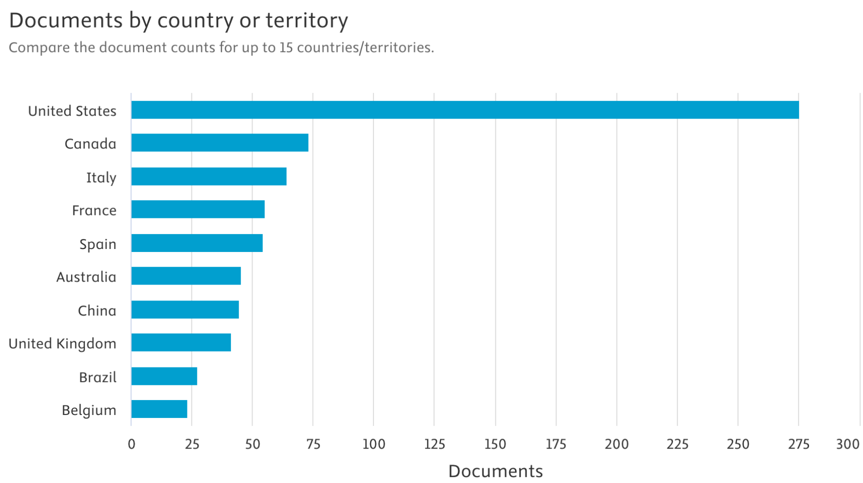


Figure 4 - Scopus documents country overview

The country is highly relevant data for this research, as the review serves future research that will mostly rely on real datasets collected based on Portugal's reality. Still, valuable literature reinforces the knowledge on what are the baselines of diagnosis of sleep apnea conditions in children that are applied worldwide.

These results were exported to be used in a Python script that checked for duplicates.

```python
import pandas as pd

file_path = '../scopus_722.csv'
data = pd.read_csv(file_path)
duplicates = data[data.duplicated(subset='Title', keep=False)]
duplicates.to_csv('duplicates.csv', index=False)
cleaned_data = data.drop_duplicates(subset='Title', keep='first')
cleaned_data.to_csv('cleaned_scopus.csv', index=False)

print(f"Found {len(duplicates)} duplicates.")
```

Figure 5 - Python script to find duplicates

The script found 2 duplicates of the 722 results.

Table 2 - Scopus search duplicates

Author	Title	Year	Source
Zampoli M.; Abousetta N.; Vanker A.	Overnight oximetry as a screening tool for moderate to severe obstructive sleep apnoea in South African children	2019	South African Medical Journal
Zampoli M.; Abousetta N.; Vanker A.	Overnight oximetry as a screening tool for moderate to severe obstructive sleep apnoea in South African children	2018	South African medical journal = Suid-Afrikaanse tydskrif vir geneeskunde

The second query, with Machine Learning, filtered 18 documents with information that reinforces the dependency on known traditional diagnosis methods but also highlights studies on the application of specific algorithms to detect different anomalies, like Cloud algorithm-driven oximetry-based diagnosis [10], XGBoost [11], logistic regression, support vector machine, AdaBoost and MAML algorithms. There are some highlighting study results from the application of deep learning to improve the diagnostic capacity of oximetry in the detection of pediatric OSA.

This search only found documents between 2018 and 2024, even though the query limits were from 2014 to 2025.

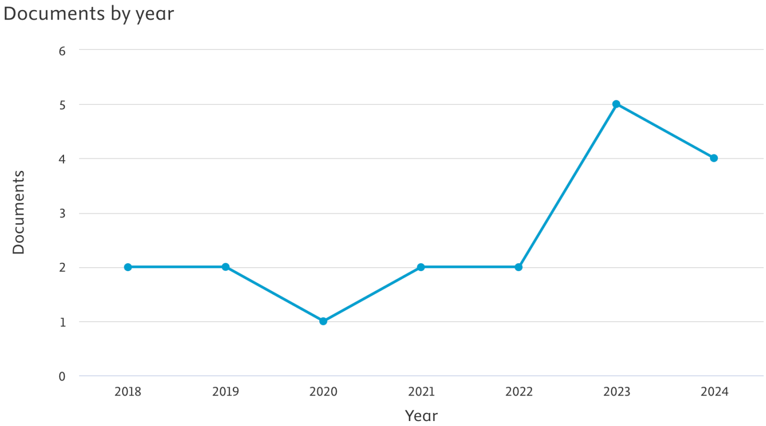


Figure 6 - Scopus search with ML year overview

For this query, we can see a pick in 2023, with five documents found, which also reinforces the increase in documents in the last 2 years.

Web Of Science

This search filtered 1.194.410 results based on articles, review articles, and papers in English or Portuguese covering the time interval from 2014 to 2025.

Only 2.298 references to sleep apnea and polysomnography were found. Of those, only 184 were related to subjects under the pediatric scope.

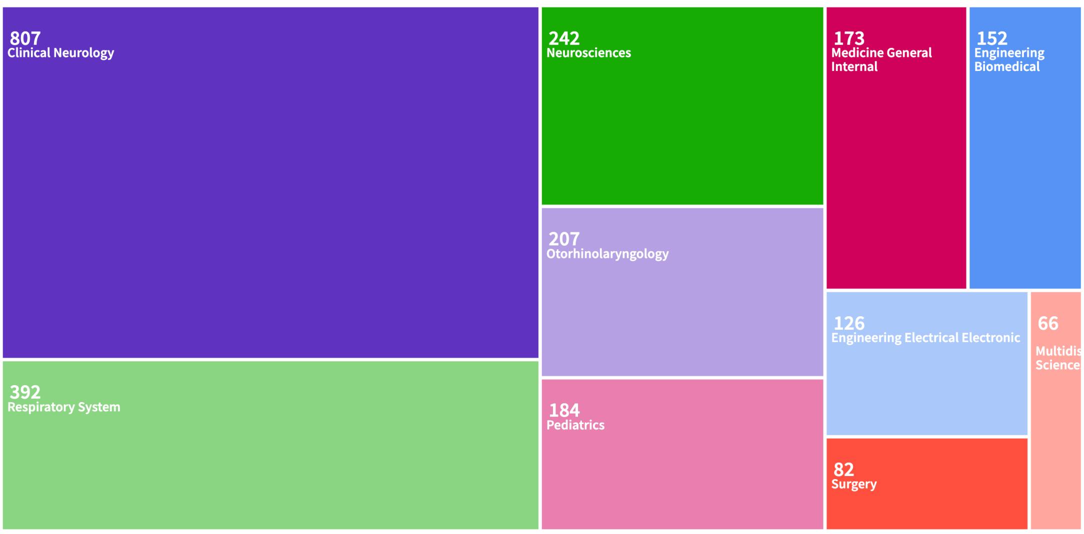


Figure 7 - WebOfScience subjects and pediatrics

A script to check for duplicates was also created for the exported results of this query.

```python
import pyexcel as pe
import pandas as pd

data = pe.get_sheet(file_name='../webofscience_191.xls')
data_frame = pd.DataFrame(data.to_array(), columns=data.row[0])
data_frame = data_frame[1:]
duplicates = data_frame[data_frame.duplicated(subset='Article Title', keep=False)]
duplicates.to_excel('duplicates_found.xlsx', index=False)

print(f"Found {len(duplicates)} duplicates.")
```

Figure 8 - Python script for duplicates in .xls files

The script did not find any duplicates from these results.

Combined Results

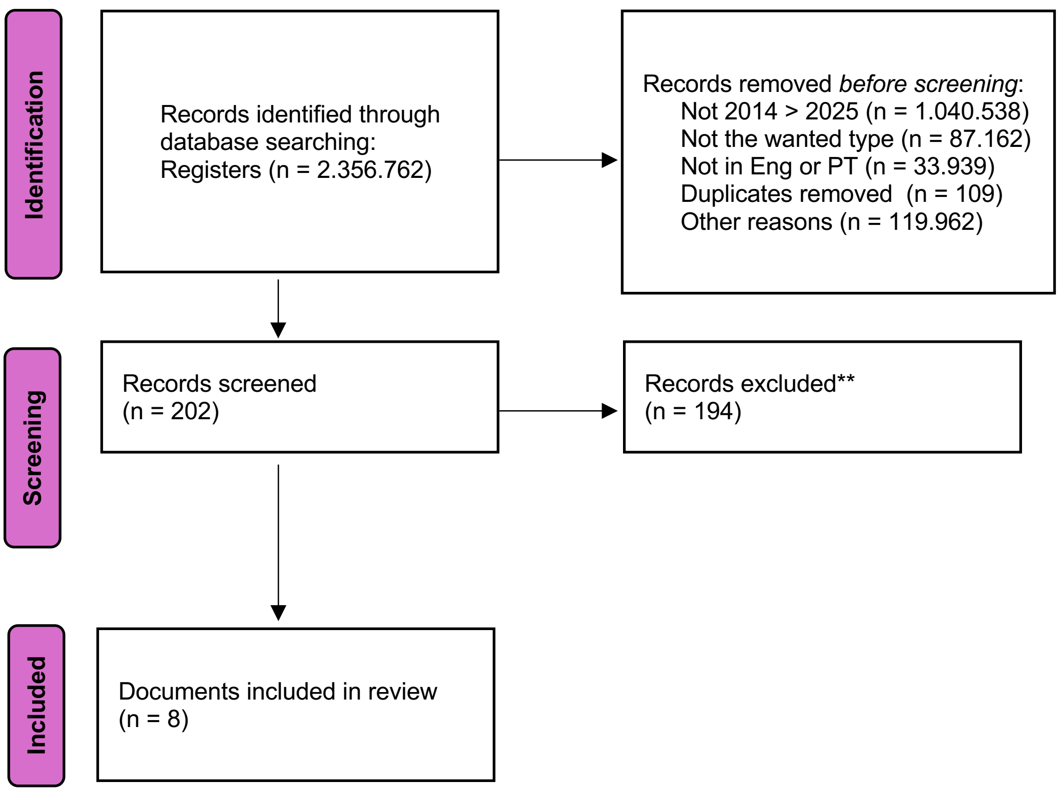
In the end, a merge from both exports was performed, relying on a created Python script.

```python
import pandas as pd

xls_file_path = '../webofscience_191.xls'
xls_data = pd.read_excel(xls_file_path, engine='xlrd')
csv_file_path = '../scopus_722.csv'
csv_data = pd.read_csv(csv_file_path)
merged_data = pd.merge(xls_data, csv_data, left_on='Article Title', right_on='Title', how='outer')
merged_data.to_csv('merged_file.csv', index=False)

print("Files merged successfully! Saved as 'merged_file.csv'.")
```

Figure 9 - Python script to merge exports

After merging, the check for duplicates script was applied, and 109 duplicates were found.

PRISMA Flow Diagram for the extration overview:



Details of the included documents:

Table 3 - Final considered results

Authors	Title	Year	Source	Type
Qin H.; Zhang L.; Li X.; Xu Z.; Zhang J.; Wang S.; Zheng L.; Ji T.; Mei L.; Kong Y.; Jia X.; Lei Y.; Qi Y.; Ji J.; Ni X.; Wang Q.; Tai J.	Pediatric obstructive sleep apnea diagnosis: leveraging machine learning with linear discriminant analysis	2024	Frontiers in Pediatrics	Article
Gutiérrez-Tobal G.C.; Álvarez D.; Kheirandish-Gozal L.; del Campo F.; Gozal D.; Hornero R.	Reliability of machine learning to diagnose pediatric obstructive sleep apnea: Systematic review and meta-analysis	2022	Pediatric Pulmonology	Article
Kim T.; Kim J.-W.; Lee K.	Detection of sleep disordered breathing severity using acoustic biomarker and machine learning techniques	2018	BioMedical Engineering Online	Article
Li J.; Sun M.; Zhao Z.; Li X.; Li G.; Wu C.; Qian K.; Hu B.; Yamamoto Y.; Schuller B.W.	Battling with the low-resource condition for snore sound recognition: introducing a meta-learning strategy	2023	Eurasip Journal on Audio, Speech, and Music Processing	Article
Ye P.; Qin H.; Zhan X.; Wang Z.; Liu C.; Song B.; Kong Y.; Jia X.; Qi Y.; Ji J.; Chang L.; Ni X.; Tai J.	Diagnosis of obstructive sleep apnea in children based on the XGBoost algorithm using nocturnal heart rate and blood oxygen feature	2023	American Journal of Otolaryngology - Head and Neck Medicine and Surgery	Article
Calderón J.M.; Álvarez-Pitti J.; Cuenca I.; Ponce F.; Redon P.	Development of a minimally invasive screening tool to identify obese Pediatric population at risk of obstructive sleep Apnea/Hypopnea syndrome	2020	Bioengineering	Article
Vaquerizo-Villar F.; Alvarez D.; Kheirandish-Gozal L.; Gutierrez-Tobal G.C.; Barroso-Garcia V.; Santamaria-Vazquez E.; Campo F.D.; Gozal D.; Hornero R.	A Convolutional Neural Network Architecture to Enhance Oximetry Ability to Diagnose Pediatric Obstructive Sleep Apnea	2021	IEEE Journal of Biomedical and Health Informatics	Article
Xu Z.; Gutiérrez-Tobal G.C.; Wu Y.; Kheirandish-Gozal L.; Ni X.; Hornero R.; Gozal D.	Cloud algorithm-driven oximetry-based diagnosis of obstructive sleep apnoea in symptomatic habitually snoring children	2019	European Respiratory Journal	Article

3.3 DSRM for Model Development

DSRM was employed to design, develop, and evaluate an ML-based diagnostic framework tailored to the challenges identified in the literature review.

3.3.1 Problem Identification

Insights from PRISMA-guided literature review and expert interviews aim to identify gaps in traditional diagnostic approaches, such as high costs, limited availability of PSG and the key points where automation can support clinicians on end diagnosis.

3.3.2 Solution Design

The proposed framework integrates patient history and PSG data to train ML models. Initial prototypes were developed using supervised learning algorithms to automate diagnostic processes.

3.3.3 Evaluation and Refinement

Evaluation of the models relied on metrics such as sensitivity, specificity, and precision-recall curves. Feedback from clinical experts informed iterative refinements.

3.4 Expert Interviews

Qualitative insights from three experts, such as Joana Marques, a PhD nurse specializing in pediatric sleep disorders, and two expert physicians from two leading hospitals, validated the traditional clinical record methods and the PSG process. In addition, these interviews validated the practical relevance of the ML framework and informed its design.

4. Findings

Besides the different focus of objective of the search resulting documents, key statements that this research is focus are ensured, such as:

Table 4 - Findings overview

Statement	Observation
Polysomnography accessability constrains.	Several documents reinforce how demanding PSG exams can be in terms of costs and infrastructure.
Diagnosis based on SCR only can become fragile.	Several documents state clinicians rely on PSG as a gold standard to support clinical patient reports.
Sleep apnea impact on congnitive development.	Several references also mention how untreated sleep apnea can affect overall child mental and emotional development.
Wearables can help support diagnosis confidence.	References exist to how oxygen and activity monitoring wearables can highlight problems and lead patients to health professionals for an early diagnosis.
Machine Learning algorithms can help increase confidence on final diagnosis.	Was possible to find documents that hightlight valuable results that support this statement.

These are only the key topics on which this research focuses on to understand how the application of machine learning algorithms can help clinicians diagnose with confidence by combining clinical patient analysis and reports with polysomnography results.

4.1 Current Diagnostic Approaches

Clinical evaluations, including SCR and PSG, remain the cornerstone of pediatric sleep apnea diagnosis. However, studies highlight significant variability in clinician assessments and diagnostic outcomes, emphasizing the need for standardized methods.

4.2 ML Applications

ML models have demonstrated promising results in automating the analysis of PSG data, detecting apnea events, and stratifying patients by risk levels. Supervised learning algorithms, such as random forests and support vector machines, are commonly used, while deep learning approaches, including convolutional neural networks (CNNs), show potential for handling complex time-series data from PSG.

5. Discussion

Integrating ML into pediatric sleep apnea diagnosis offers significant opportunities but is not challenge-free. While ML may improve clinicians' diagnostic confidence levels, ethical considerations, including data privacy and accessibility, are also critical.

Future research should develop robust, interpretable models integrating diverse data sources, such as PSG, patient history, and wearable device outputs. Collaboration between technology experts and healthcare providers is essential to ensure these tools are practical and clinically relevant.

6. Conclusion

The application of machine learning techniques to pediatric sleep apnea diagnosis stands at the crossroads of traditional diagnostic methods and cutting-edge technological innovation. This review highlights the studies conducted on such technologies within the last two years window. This reinforces the need to continue this research to exploit this momentum.

When profoundly diving into the different constraints and complexity levels that characterize a polysomnography exam and the awareness that one has of the advancements in electronics, especially how accurate small chips and sensors can be today, it almost points to what seems to be the apparent priority that should be on reducing the existent complexity of such exams. This review unveiled several studies that reinforce this fact. However, this research will continue to consider the existing methods and resulting datasets, hopping to specific variables and details that other researchers can use to improve PSG comfort levels to make it easier to apply to children.

These challenges highlight the urgent need for more accessible and efficient diagnostic solutions but also support the resulting data analysis to reduce the need to repeat the exam. Machine learning presents a transformative opportunity by offering the ability to analyze vast and complex datasets, automate processes, and integrate multiple data sources such as patient history and polysomnographic results.

The findings of this review also underscore the potential of ML to enhance diagnostic accuracy, reduce diagnostic delays, and make advanced diagnostics more accessible in resource-limited settings. By automating key aspects of the diagnostic workflow, ML tools alleviate the burden on clinicians and enable early and accurate identification of pediatric sleep apnea, critical to mitigating its long-term cognitive, behavioural, and physiological impacts.

However, successfully integrating ML into clinical practice requires addressing several challenges. These include ensuring the availability of high-quality and diverse datasets to train robust models, managing ethical considerations around data privacy, and fostering trust and interpretability of ML outputs among clinicians.

The focus is far from completely automating OSA diagnosis processes; instead, it is to serve future research and maintain the continuous improvement of a subject that has seemed frozen in time for more than twenty years. Future research should focus on refining ML algorithms to handle the inherent variability in pediatric populations, exploring wearable devices for continuous monitoring, and developing user-friendly platforms to facilitate clinician adoption. Additionally, longitudinal studies that evaluate the real-world impact of ML-based diagnostic tools on patient outcomes will be vital to demonstrating their clinical efficacy and cost-effectiveness.

By combining traditional diagnostic methods' strengths with machine learning's innovative capabilities, the field of pediatric sleep medicine is poised for significant advancements. This integration promises to improve diagnostic precision, optimise treatment pathways, and ultimately enhance the quality of life for children suffering from sleep disorders. Pediatric sleep apnea diagnosis is at the crossroads of traditional methods and technological innovation. Machine learning can potentially transform this field by addressing the limitations of current approaches and enhancing diagnostic workflows. Continued research and interdisciplinary collaboration are needed to realize the full potential of ML in improving outcomes for children with sleep disorders.

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